Meeting with S.R.I. people Feb 13, '58

Dimwell: "STRETCH"

2½ yrs. — 2 yrs to go

(1) components:

drift transistors: 10 Megapulse opns. { switching times

— Speeds are actually here. — hard to push higher.

new memory: 0.5 μs full cycle — multi-hole ferrite

— now 10[illegible] at 0.6 μs — better bargain.

— Transit time is limit to go to large mem.

addressing ½ million words.

— now 16K 2.2 μs — toroid.

(2) planning objectives

(1) Technical & commercial use

— more effort can be put into

(2) 100 times performance (rate) a.u. flpt. and logical.

100 × useful work

— multiplexing —

variable field, variable record, binary, code with any no. of bits; convert from code to code by mapping.

— Data transmission

(3) I/O system would be general — not prejudiced for or against any form of data.

(eg. character recognition)

with unit Several units simultaneously. — Telephone lines

(Q: pulse shapes? will be "moderately choosy" with standards — matching.)

| Serial A.U: all logical, do bit wise

Arith unit ← single format, high speed 48 bit mantissa

word size in mem. 64 bits, can cross boundaries.

(4) program interrupt system — regular
— can write prog. regardless of interrupts

example: telephone inquiry for file in Remote disk.

— levels of interruption? 1, 2, 3 ...
— priorities

Business appl?— of future — machine will be "in line" in real time
incomplete data must be "cleaned up" — more work.
— character recognition

Philosophy of operation? : — a machine is well-loaded now.
how can person tell if he is going to
get on such a machine — priority —?

can't have but one primary problem at a time & that
is not dominated by I/O Times.

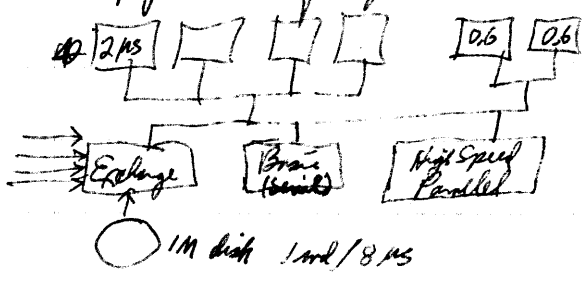
Q: Set of requirements on business machine — is there a group
looking into — A: "Integrated Data processing" group in research.
Nature of business — tool effects the solving of problem.
systems study

Q: what business can afford it? A. Treasury uses 7.25 ¢.
cut costs in half for process, errors were cut

Q: components, no of.? A. will vary 100,000 transistors up.

Q: Arith speeds A. goals 0.6 Add 1.2 Mpy 1.8 Div (+0.2 μs)
may not meet but will come close

Q: Configuration of System:



Q, Core switching A: in part.

Q: work done here? A: yes

Q, ~~Q~~ yield rates on materials? A, last word not in.

Maintenance & Reliability:

100 bits of info. from exchange

Q. Marginal checking? A. not sure it is significant,

Case: analysis & Computation Groups.

(1) ~~Use~~ job shop

(2) logic minimization. — speed — boolean algebra —

(3) logic implementation;
4 level logic.